Propagation of Laser Light: Gaussian Beam Optics

PC2193: Experimental Physics 1 Presentation

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Introduction: Motivation and Key Objective

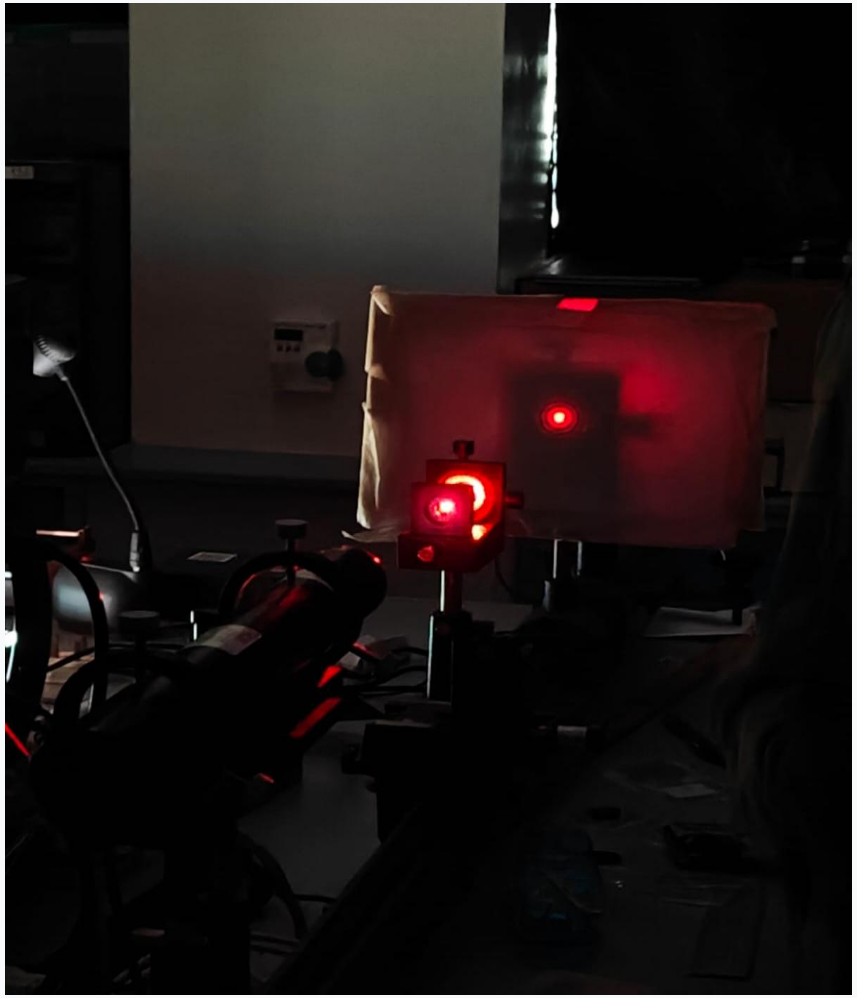
Motivation

- The invention of lasers revolutionised science, and today lasers are used in several different industries, such as medicine, microprocessors and so on.

- Lasers are often used in settings that demand high precision, so we need to know the characteristics of the laser beam.

Objectives

- This experiment studies and verifies the Gaussian intensity profile of the given laser by obtaining the M^2 value. We also calculate the beam waist (W_0) and the Rayleigh Range(Z_R) before obtaining the M^2 value.



Laser being used in the study of diffraction (taken in PC2193 lab)

Introduction: Underlying Physics

- The intensity of the laser beam follows a Gaussian profile given by the equation:

$$I = I_0 \exp\left(\frac{-2r^2}{W^2(z)}\right) \qquad (1)$$

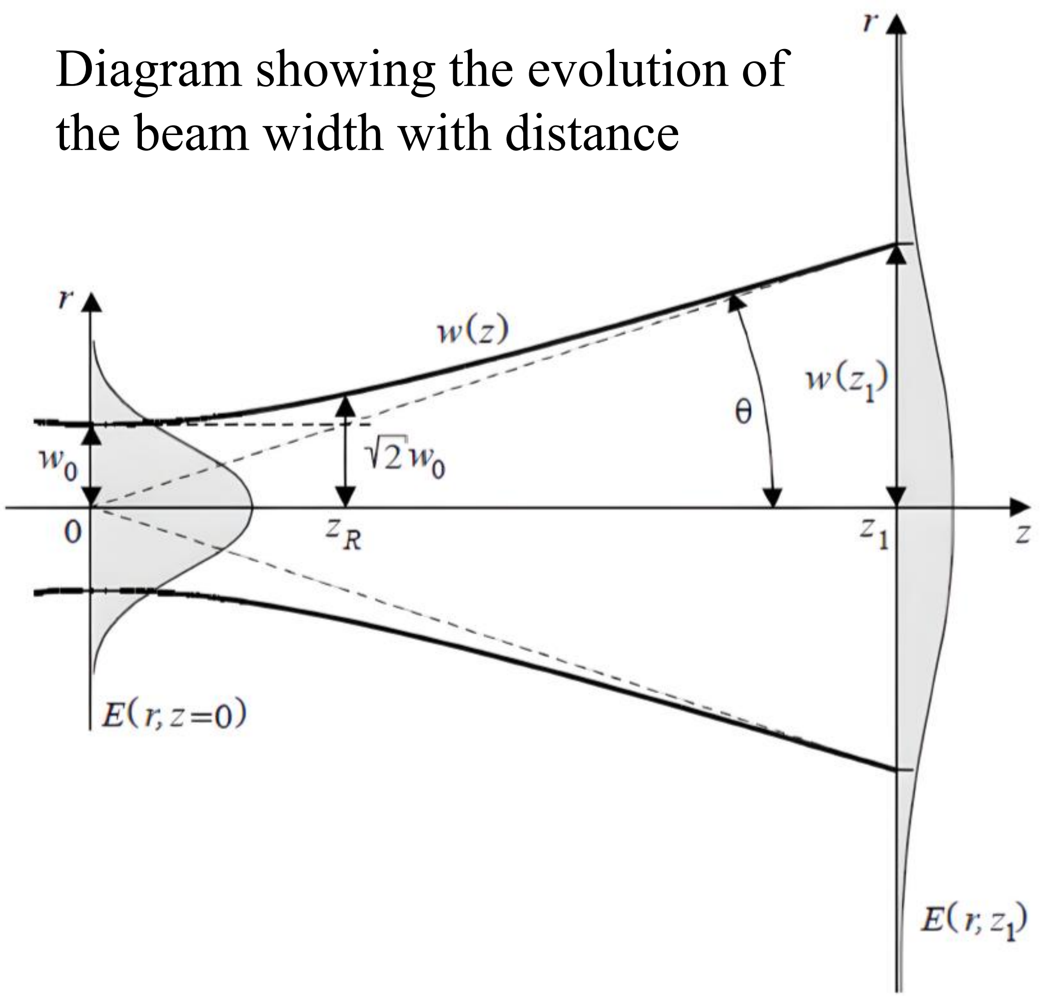
Where r is the distance from the beam axis and W(z) is the beam width or the spot size. The beam width evolves as:

$$W(z) = W_0 \sqrt{1 + \left(\frac{z}{Z_R}\right)^2} \qquad (2)$$

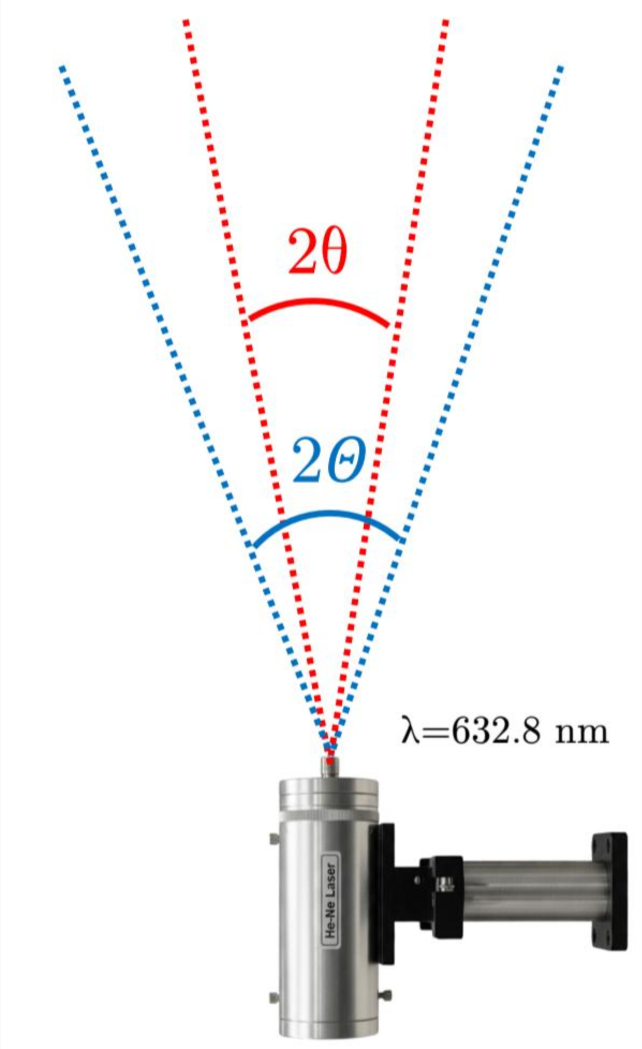
W_0 is the beam waist. Z_R is the Rayleigh range and is given by:

$$Z_R = \frac{\pi W_0^2}{\lambda} \qquad (3)$$

Diagram showing the evolution of the beam width with distance



Introduction: Underlying Physics Continued



However, in practice, the observed divergence is greater than the theoretical divergence. The M^2 parameter is a measure of this.

$$M^2 = \frac{\Theta}{\theta} > 1 \tag{4}$$

Where,

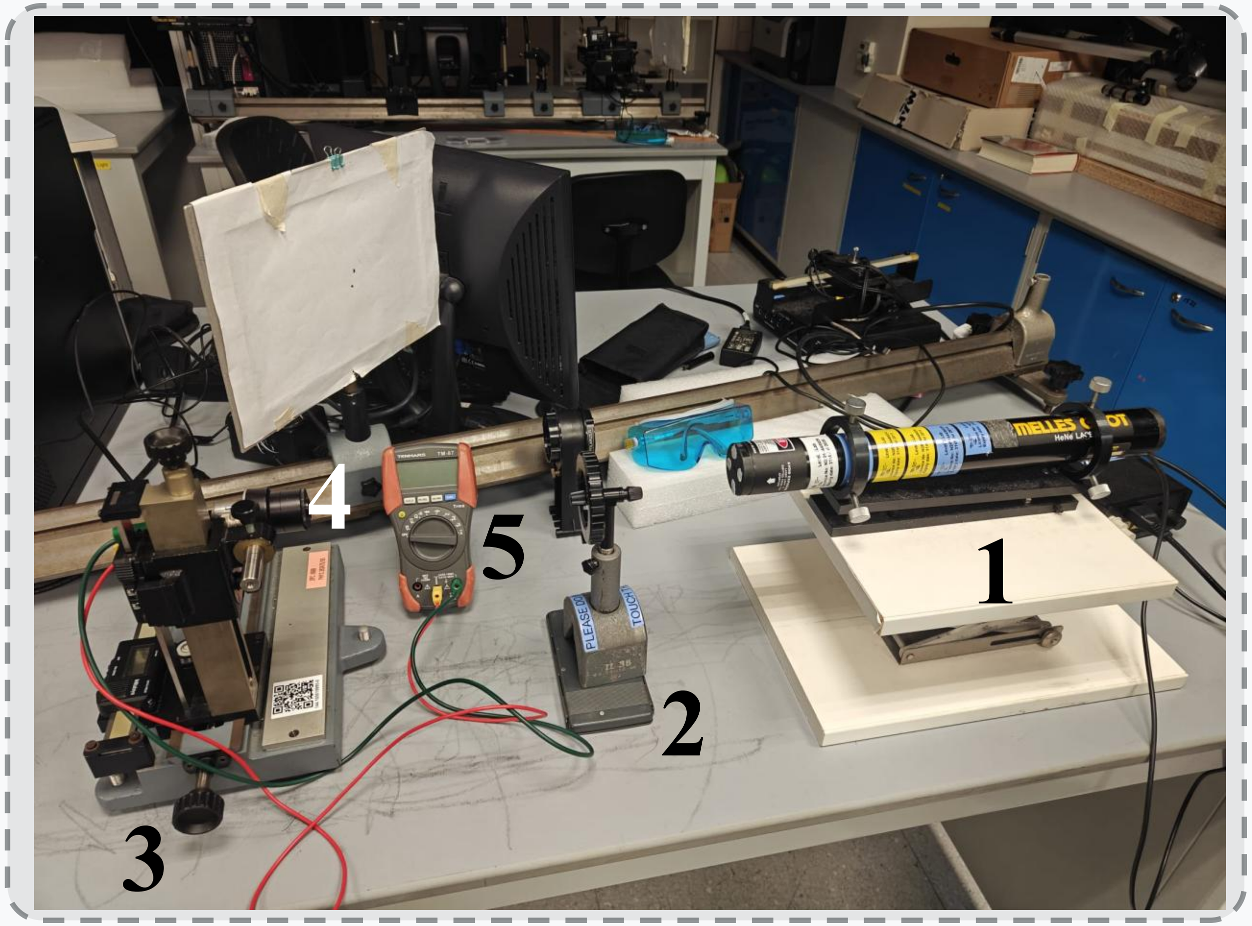
$$\theta \approx \frac{\lambda}{\pi W_0} \tag{5}$$

And, Θ is determined experimentally.

There are several reasons for M^2 always being greater than 1:

- Diffraction
- Laser profile
- Optical aberrations

Experimental Setup & Procedure



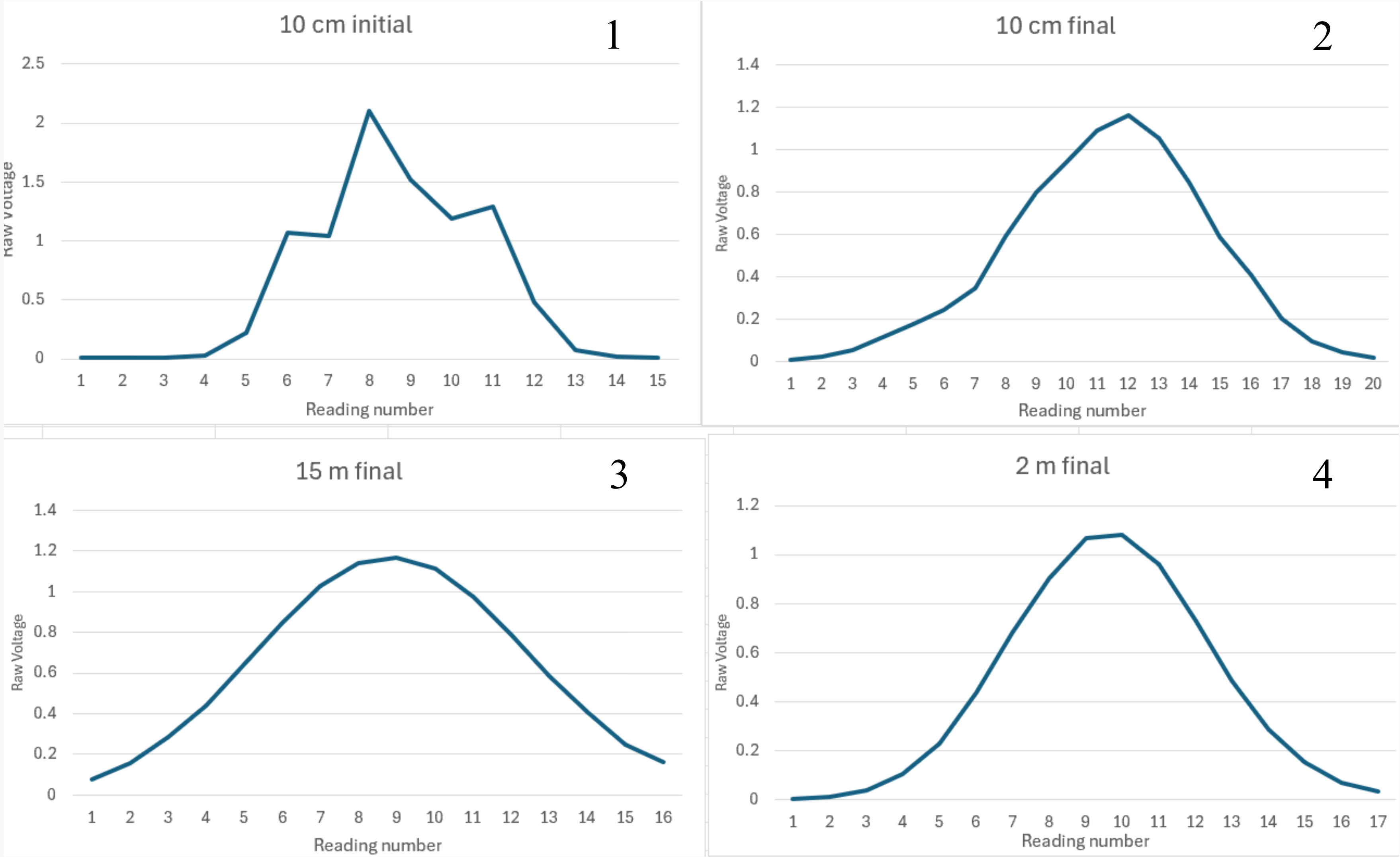
Core Components:

1. He:Ne Laser ($\lambda = 632.8$ nm)
2. Optical Attenuator
3. Micrometre Translation Stage
4. Photodetector
5. Digital Multimeter

Procedure:

1. Align the laser and photodetector
2. Adjust the attenuator accordingly
3. Set translation step
4. Adjust the micrometre stage
5. Note the micrometre readings and the corresponding voltage reading on the multimeter.

Results: Raw Data



Raw ‘sanity check’ graphs of just the voltage readings were plotted after every run.

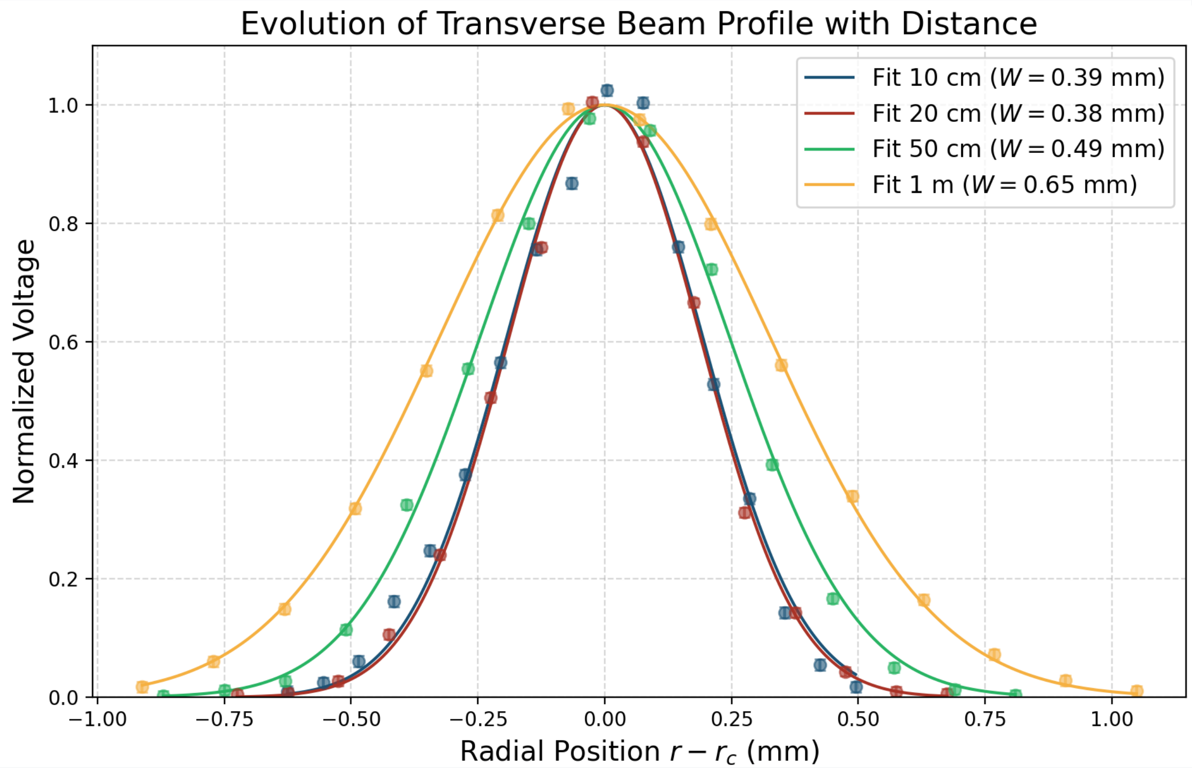
Procedure was changed when erroneous results, such as subplot 1, were obtained.

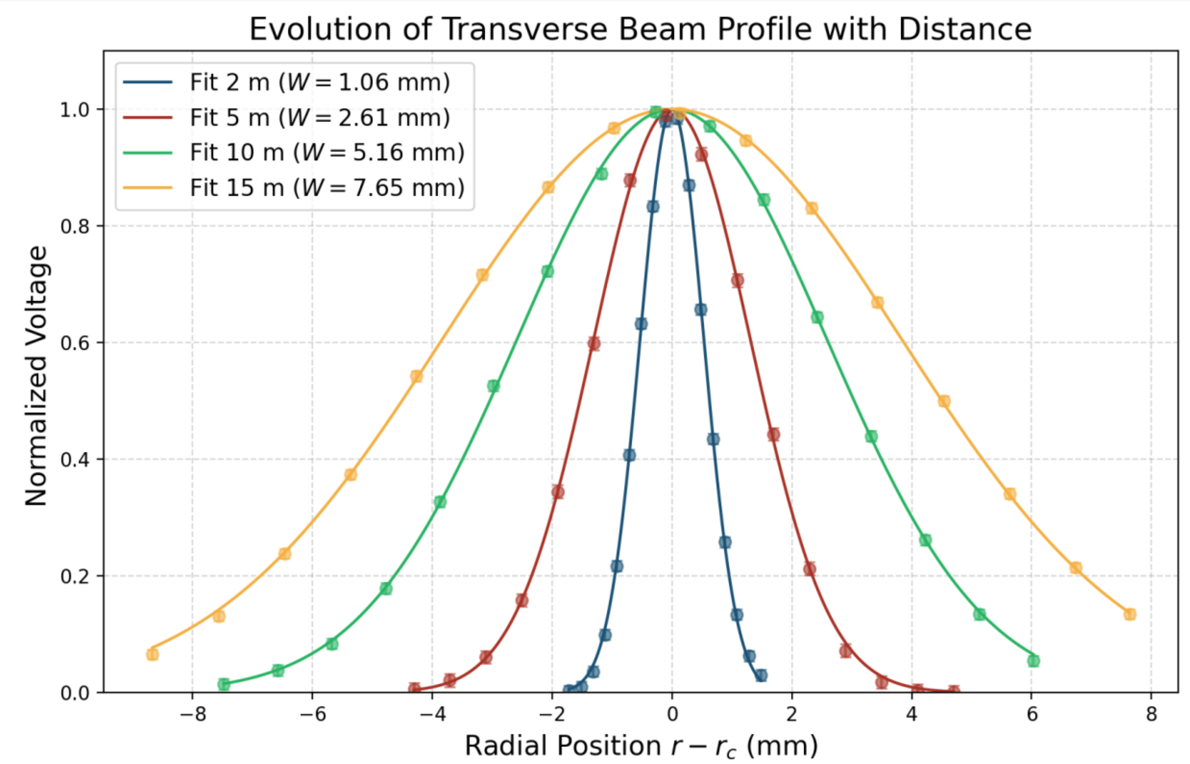
Results: Error Analysis

- **Micrometre Resolution:** ±0.01 mm (fundamental limit of the translation stage). Additionally, changes in voltage readings were observable even without a change in micrometre readings near the peak.
- **Voltage Reading Fluctuations:** Near the peak, voltage readings would fluctuate by ± 0.01 V.
- **Gaussian Fit Errors:** Residuals from the curve fitting process provided the standard error for beam waist W_0, which was = 0.0101 mm
- **Distance measurement using tape:** 0.05 m, at longer distances (2m to 15m), errors may be introduced due to misalignment, parallax and systematic errors in the tape itself.
- Error propagation in quadrature has been used throughout for absolute as well as relative errors

Results: Normalised Plots and Fitting

Plots of normalised voltage vs radial position with error bars and fitted curves





The voltage readings were normalised, and curve fitting was done using a Gaussian fitting function based on equation (1). The beam width W(z) was left as a free parameter. NumPy and SciPy were used.

Results: Beam Waist and Rayleigh Range

At close distances (10 cm), $W(z) \approx W_0$. Thus, the determined value of beam width at 10cm is taken as W_0.

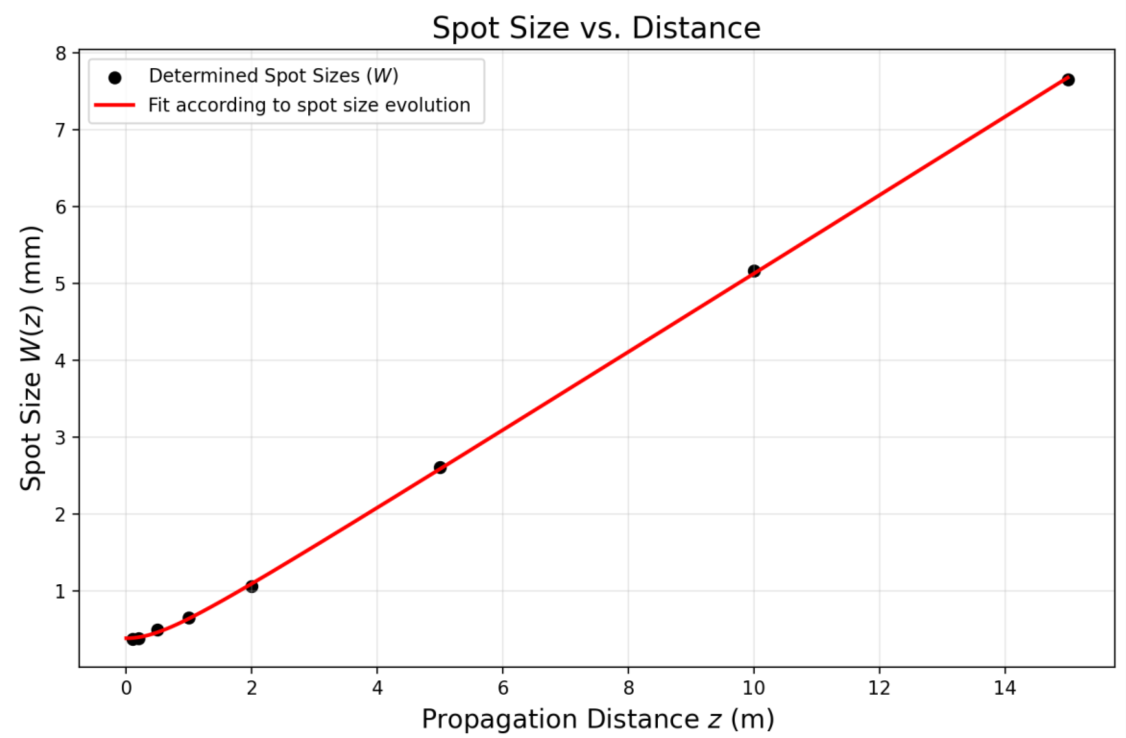
$$W_0 \approx W(0.1m) = 0.39 \ \pm 0.014 \ mm$$

Using this value of W_0, the Rayleigh range is determined using equation (3).

$$Z_R = \frac{\pi \times (0.39 \times 10^{-3})^{\wedge}2}{632.8 \times 10^{-9}} = 0.755 \pm 0.054 \text{ m}$$

θ is determined using equation (4)

$$\theta = \frac{632.8 \times 10^{(-9)}}{\pi \times 0.39 \times 10^{(-3)}} = \ 0.516 \ \pm 0.019 \text{ mrad}$$



To confirm our initial assumption, a non-linear regression was performed to fit spot size vs. propagation distance. W_0 obtained this way was 0.387 mm, which is very close to our assumption.

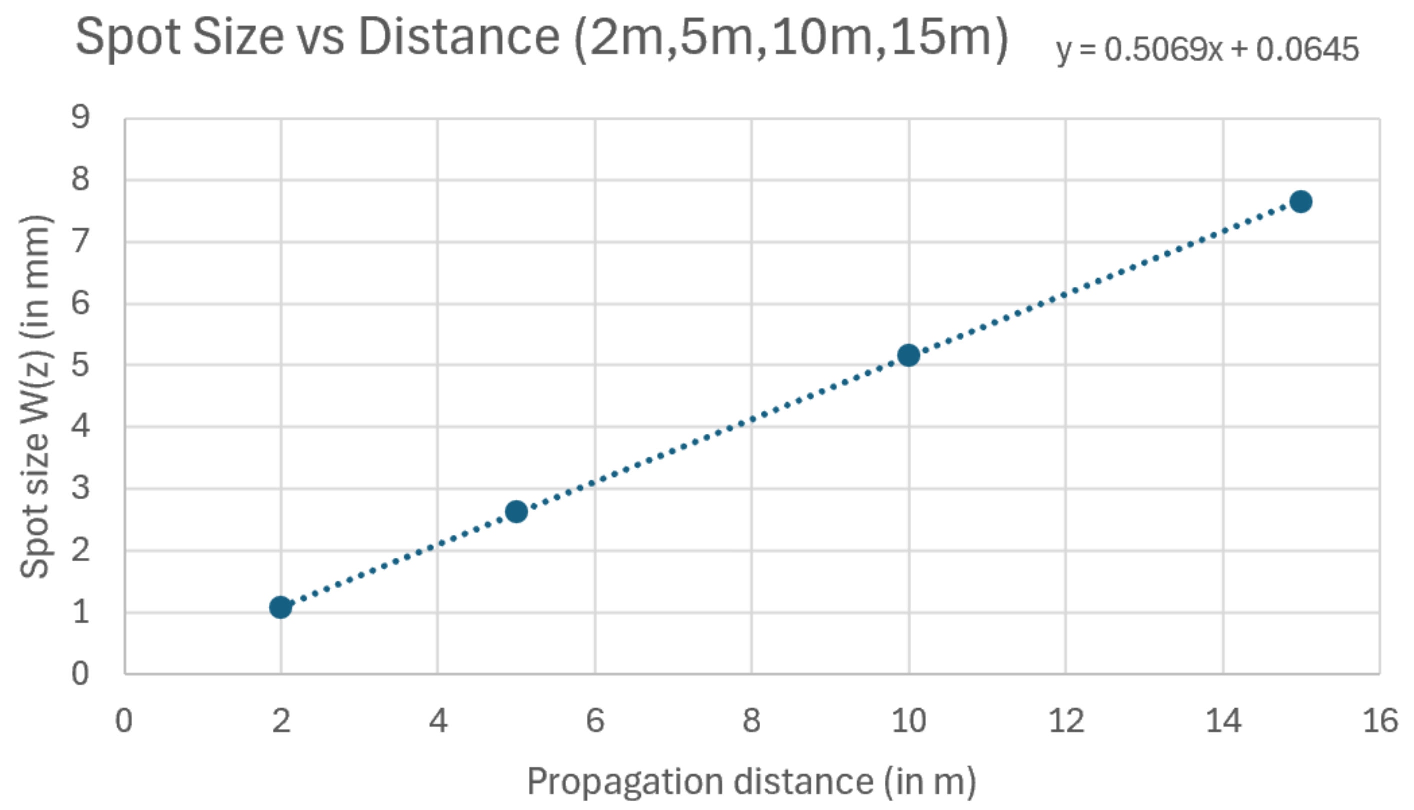
Results: Θ and M² values

Θ is determined by considering the slope of spot size vs long propagation distances.

$$\Theta = 0.507 \pm 0.003 \text{ mrad}$$

Finally, we can calculate M^2 using equation (4).

$$M^2 = \frac{0.507}{0.516} = 0.983 \pm 0.036$$



Graph of spot size vs propagation distance with the lower distances (0.1 m – 1m) removed.

Discussion: Validity, Error Sources & an Interesting Observation!

Validity : The experimentally obtained value of M^2 (0.983 ± 0.036) deviates from the ideal value (=1) by 1.7% and is less than 1, which is physically impossible. However, the uncertainty bounds suggest that we should not disregard the result since the upper bound value of 1.019 is possible. This result confirms that the He:Ne laser is an exceptionally high-quality source, operating near the diffraction limit with less than 2% deviation from the ideal Gaussian model. Nevertheless, there are several likely systemic errors that may have led to the value being lower than 1.

Sources of Uncertainty

- Micrometre resolution limits
- Low number of translation steps at shorter distances
- Beam wandering (thermal drift)
- Parallax during manual reading
- (Interesting observation) Deviation in the table due to force applied.

Conclusion

Parameter	Measured Value ± Uncertainty
Beam Waist (W_0)	0.39 ± 0.014 mm
Rayleigh Range (Z_R)	0.755 ± 0.054 m
Beam Quality (M^2)	0.983 ± 0.036

The He:Ne laser demonstrates near-ideal Gaussian propagation with a beam quality factor M^2 close to unity, validating the TEM_0 mode assumption for this experimental setup. M^2 value below 1 points to systemic errors, particularly for the shorter distances (10cm, 20cm). Future iterations can be improved by increasing translation steps at shorter distances, employing a micrometre with better accuracy and making sure to minimise random error as much as possible by letting the laser reach steady-state and not putting force on the table.